Name …….

1. An unknown semiconductor has Eg = 1.1 eV and $Nc = Nv$. It is doped with 10^{16} cm^{-3} donors, where the donor level is 0.24 eV below Ec. Given that E_f is 0.29eV below Ec. Calculate intrinsic concentration, n_i, and the concentration of electrons and holes in the semiconductor at 300 K.

2. A Si sample is doped with 10^{17} boron atoms/cm3.

i. What is the electron concentration n_0 at 300 K?
ii. What is the conductivity?
ii. What is the resistivity?

[Mobility of holes, μ_p = 250cm^2/V-s]

3. 2V is applied across a 1um piece of n-type silicon device. The doping concentration is N_D = 10^{17} cm^{-3}

(i) If the transit time for an electron is 4ps. Calculate the average drift velocity of an electron in the device

(ii) What is the electron mobility?

(iii) Calculate the electron diffusion coefficient

(iv) Determine the diffusion current at the midpoint of the bar

(v) Determine the drift current

4. A Si p-n junction has Na = 10^{17} cm^{-3} on one side and N_d=2 x 10^{15} cm^{-3} on the other.
(a) Calculate the Fermi level positions at 300 K in the p and n regions.
(b) Draw an equilibrium band diagram for the junction and determine the contact potential V_0 from the diagram.
Find the following;
(c) Depletion width, W.

[$n_i = 1.5 \times 10^{10} cm^{-3}$; $\varepsilon_0 = 8.85 \times 10^{-14}\ F/cm$ $\varepsilon_s = 11.8$]